

MAN00035C



**School for Business
and Society**

Quantitative Analysis in Business Studies

2026 - Week 6

Week 6: The Chi-squared Test (χ^2 Test)

Reading: Oakshott Chapter 9

Last topic

We look at

- Null hypotheses and alternate hypotheses
- Critical values and p-value
- Type I and Type II error
- Test statistics for means, proportions, “paired data” and two samples.

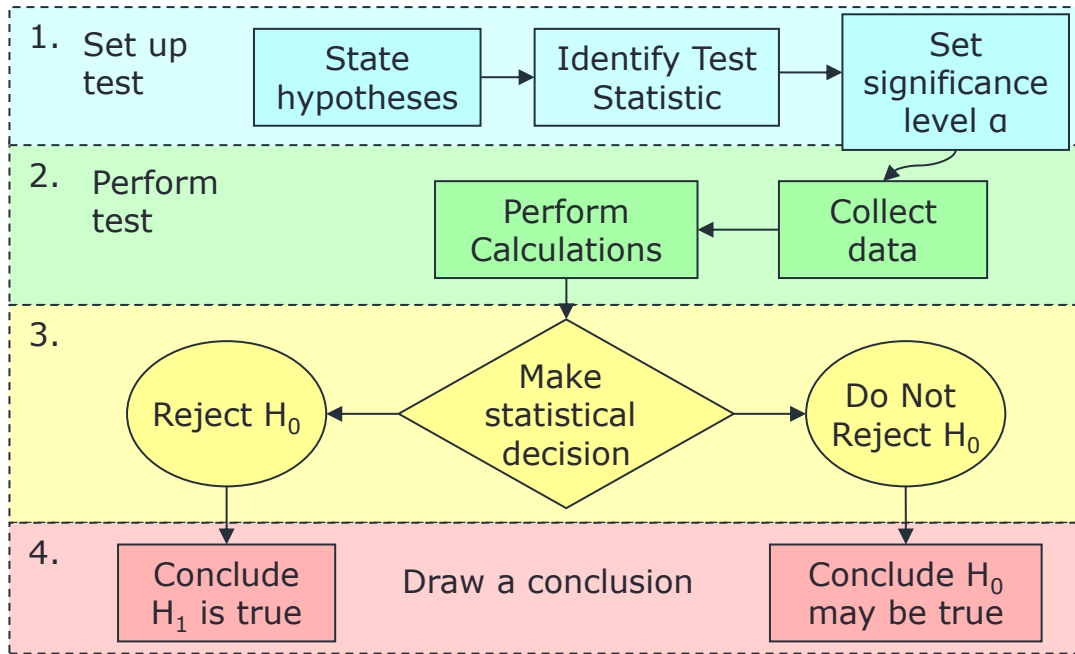
This topic

We looked at:

- Contingency table
- Chi square test
 - **Goodness of fit test**
 - **Test of Association**

From the last lecture:

Steps for carrying out a hypothesis test



Step 1. Set up the null and alternative hypotheses

Step 2. Decide on the significance level (normally, 5%), null distribution, and determine the critical values (**which one?**)

Step 3. Calculate the test statistic (**how?**)

Step 4. Decide whether to accept or reject the null hypothesis. (**how?**)

Step 5. Draw a conclusion.

Refresher

- What types of data do we have? (Week 1)
 - Categorical
 - Scalar
- What types of test did we discuss last week?
 - Scalar: test the mean
 - Dichotomous: test the proportion
 - Scalar vs. Dichotomous: test the difference between two sample means
- What if the data is Categorical?
 - Categorical
 - Categorical v.s. Categorical (Dichotomous v.s. Dichotomous)
 - Scalar vs. Categorical (Next week)

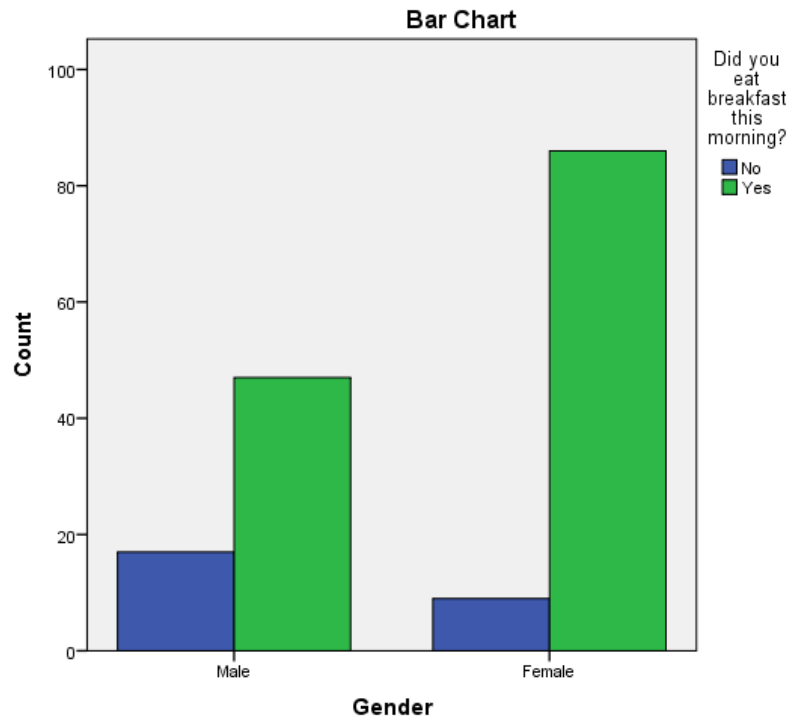
Gender ▾	Height (m) ▾	Weight (kg) ▾	Height Guess ▾	Weight Guess ▾	Breakfast? ▾	College ▾	Confident With Numbers ▾
Female	1.52	46	1.52	77	Yes	Goodricke	Agree
Male	1.72	76	1.52	66	Yes	Derwent	Agree
Female	1.52	40	1.53	59	Yes	Goodricke	Agree
Male	1.65	61	1.54	57	No	Langwith	Disagree
Male	1.8	72	1.54	77	Yes	Langwith	Agree
Female	1.54	45	1.57	61	Yes	Derwent	Disagree
Male	1.69	69	1.57	70	Yes	Derwent	Agree
Female	1.47	25	1.58	72	Yes	Constantine	Disagree
Female	1.83	69	1.58	61	Yes	James	Strongly Agree
Female	1.81	61	1.6	52	Yes	Constantine	Strongly Agree
Female	1.7	68	1.61	74	Yes	Derwent	Agree
Male	1.9	84	1.61	69	Yes	James	Agree
Female	1.57	49	1.62	64	Yes	Constantine	Agree
Female	1.65	44	1.62	70	Yes	Constantine	Strongly Agree
Female	1.71	56	1.62	74	Yes	Langwith	Neither Agree nor Disagree
Female	1.72	56	1.62	77	Yes	Constantine	Neither Agree nor Disagree
Female	1.75	71	1.62	88	Yes	Langwith	Neither Agree nor Disagree
Male	1.92	83	1.62	87	No	James	Disagree
Female	1.65	48	1.63	85	Yes	Langwith	Neither Agree nor Disagree
Male	1.71	75	1.63	70	Yes	Langwith	Neither Agree nor Disagree

dichotomous vs dichotomous– example

Gender * Did you eat breakfast this morning?
Crosstabulation

Count

		Did you eat breakfast this morning?		Total
		No	Yes	
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159



Dichotomous² – example

Start with a similar null hypothesis; which of the two groups you're in does NOT affect the percentage / proportion.

Null Hypothesis: No difference. That is, $H_0: \pi_1 = \pi_2$.

Alternate Hypothesis: Some difference. $H_1: \pi_1 \neq \pi_2$.

Dichotomous² – example

For continuous variables, the mean and variance are usually determined by two parameters μ and σ . HOWEVER, that's not the case here!

(Remember the Bernoulli?)

π gives us both middle AND measure of spread.

By assuming $\pi_1 = \pi_2$, the two measures of spread are the same as well.
So that implies pooling again.

Dichotomous vs dichotomous

Oakshott gives this as

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

but it might be easier to think of it as

$$\hat{p} = \frac{\text{Sum of "wanted"}}{n_1 + n_2}$$

Then, remembering that the variance for the Bernoulli is $p(1 - p)$, we have

$$STED = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$(cf. s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right)})$$

Again, $TS = \frac{p_1 - p_2}{STED}$, and null distribution is $N(0,1)$.

Dichotomous² – example

- We might wonder whether gender is associated with the proportion of people that have breakfast:

$$H_0: \pi_1 = \pi_2.$$

$$H_1: \pi_1 \neq \pi_2.$$

$$\hat{p} = \frac{47 + 86}{64 + 95} = \frac{133}{159}$$

$$STED = \sqrt{\frac{133}{159} \left(1 - \frac{133}{159}\right) \left(\frac{1}{64} + \frac{1}{95}\right)}$$
$$= 0.0598$$

Gender * Did you eat breakfast this morning?
Crosstabulation

Count

		Did you eat breakfast this morning?		
		No	Yes	Total
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

Two independent samples Z-test on proportions



$$H_0: \pi_1 = \pi_2.$$

$$H_1: \pi_1 \neq \pi_2.$$

$$TS = \frac{47/64 - 86/95}{0.0598} = -2.857$$

CV: row $64 + 95 - 2 = 157$,

2-tails, 5%: -1.960 and 1.960

We find that $TS < \text{lower CV}$, H_0 is rejected.

It appears that the proportion DOES significantly change.

Gender * Did you eat breakfast this morning?
Crosstabulation

Count		Did you eat breakfast this morning?		Total
		No	Yes	
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

Two independent samples Z-test on proportions

Gender * Did you eat breakfast this morning?
Crosstabulation

Count

		Did you eat breakfast this morning?		Total
		No	Yes	
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

$$TS = \frac{p_1 - p_2}{STED}$$

$$STED = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

This does not work for categorical vs categorical!

The idea of χ^2 test (1)

Gender * Did you eat breakfast this morning?
Crosstabulation

Count		Did you eat breakfast this morning?		Total
		No	Yes	
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

1. Assuming the two variables are independent, what is the expected counts?
2. How are the observed counts deviates from the expected counts?
3. How to standardize them and combine them into a single value (test statistics)?

Rephrase the hypothesis for categorical:

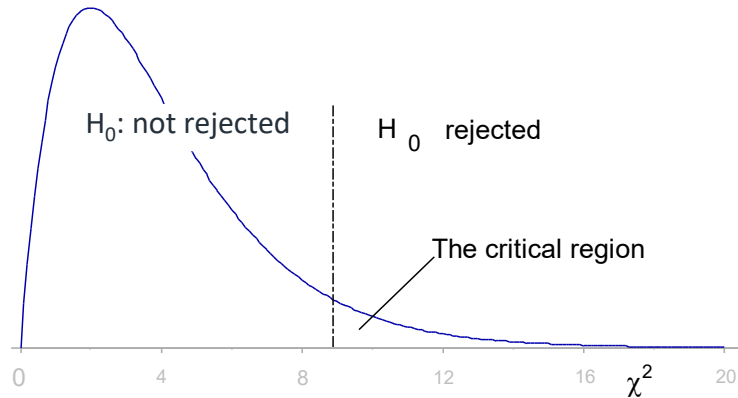
H_0 : Gender is **NOT** associated with the proportion of people that have breakfast

H_1 : Gender is associated with the proportion of people that have breakfast.

- We have been using **parametric tests** (testing a parameter, such as the mean or proportion).
- Now it is a **non-parametric test**.

The idea of χ^2 test (2)

The idea is to use the chi-square distribution, which is a continuous probability distribution that emerges when we sum squared independent standard normal random variables.



Test Statistic $\chi^2 = \sum \frac{(O - E)^2}{E}$

Degree of freedom

Gender * Did you eat breakfast this morning?
Crosstabulation

Count

		Did you eat breakfast this morning?		
		No	Yes	Total
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

In the Binary vs Binary case, $DF = 1$.

Consider the 2x2 table, if one number is known, all the other three numbers can be calculated.

Expected Counts

Gender * Did you eat breakfast this morning?
Crosstabulation

Count		Did you eat breakfast this morning?		Total
		No	Yes	
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

- The proportion of No is 26/159.
- If it is not associated with gender, among 64 males, the expected value is $64 \times 26/159 = 10.47$

Expected	No	Yes	Total
Male	10.47	53.53	64
Female	15.53	79.47	95
Total	26	133	159

TS	No	Yes	Total
Male	4.07	0.80	
Female	2.74	0.54	
Total			8.15

$$TS = \frac{(17 - 10.47)^2}{10.47} + \frac{(47 - 53.53)^2}{53.53} + 2.74 + 0.54 = 8.15$$

Equivalence of χ^2 test and Z-test for Binary vs Binary Case

- Z-test:**

Null distribution: $N(0,1)$

TS = -2.857

- χ^2 -test**

Null distribution: $\chi^2(df = 1)$

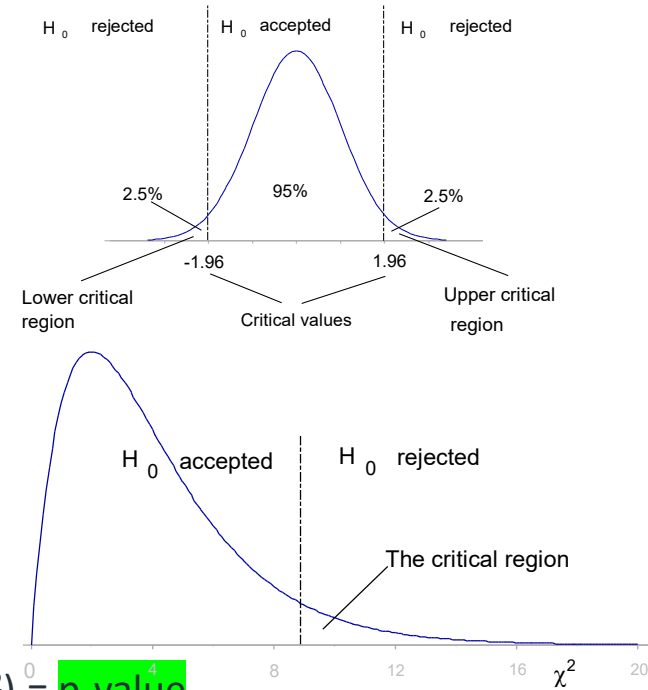
TS = 8.15

$(-2.857)^2 = 8.15$ with some rounding error

This is not coincidence but because $\chi^2(1) = Z^2$.

p-value = $P(|Z| > |-2.857|) = P(|Z|^2 > |-2.857|^2) = P(\chi^2(1) > |-2.857|^2) =$ **p-value**

Conclusion: A two-sided Z-test is equivalence to one-sided χ^2 -test with $df=1$.



Summary

Test Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

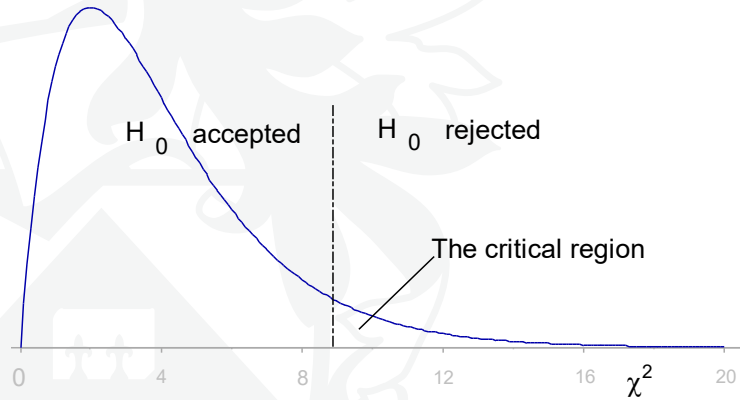
where

‘O’ is observed

‘E’ is expected value

Count the number of data items that are observed (O) in a particular category. Then find the **difference between the Observed frequency and the Expected frequency** of that category. **Square** this difference. And **divide by Expected frequency**. Repeat for all categories and **sum** the individual answers.

Summary



Test Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The Chi-Squared distribution is
a skewed distribution

The critical value is **one-sided** and
depends on the *degrees of freedom*

Where 'O' is observed

'E' is expected value

Three forms of χ^2 test

1. Goodness of fit test

- This tests to see if the data fits some distribution.
- Often investigating the distribution of one categorical (*nominal or ordinal*) variable in one sample

2. Test of Association

- Test to see if there is any association between categories in a two-way table.
- Often testing to discover an association between two categorical variables in one sample.

3. Test of Homogeneity

- Test to see if different populations or groups share the same distribution for a single categorical variable in two samples.

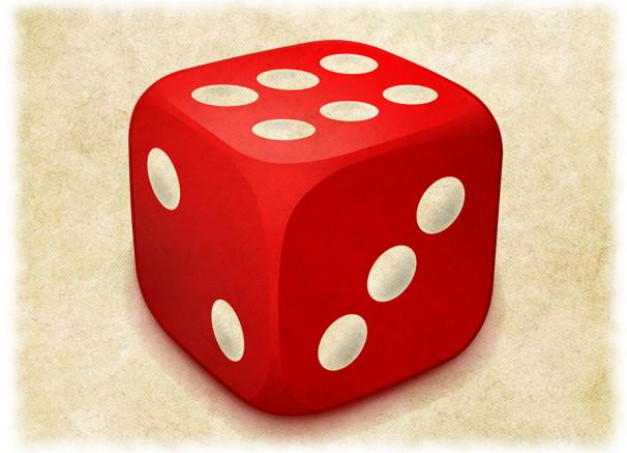
Example: throw a die

For example, throwing a die:

Number	1	2	3	4	5	6
Frequency	20	10	10	20	20	40

This is **120** throws (repetitions), so **we'd expect 20 of each number with random rolling.**

Are the frequencies obtained close enough for random rolling?



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Hypothesis

Null Hypothesis is that the die is fair.

H_0 : The distribution of die outcome is

Number	1	2	3	4	5	6
Prob	1/6	1/6	1/6	1/6	1/6	1/6

Alternate Hypothesis is that this die is Not a fair one.

H_1 : The distribution of die outcome is not uniformly distributed.

Observed vs Expected

We have Observed frequencies. We need to compare them to the **Expected frequencies**.

Number	1	2	3	4	5	6
Observed	20	10	10	20	20	40
Expected	20	20	20	20	20	20

*With a “**fair**” die, each face is equally likely to appear, therefore each is expected to appear as much as the rest.*

*With infinite rolls, would have same numbers in our **Expected** row as in the **Observed** row,*

Observed vs Expected

We have Observed frequencies. We need to compare them to the **Expected frequencies**.

Number	1	2	3	4	5	6
Observed	20	10	10	20	20	40
Expected	20	20	20	20	20	20

We then need to calculate our Test Statistic:

$$TS = \sum \frac{(O - E)^2}{E}$$

Calculating the TS

$$TS = \sum \frac{(O - E)^2}{E} = 5 + 5 + 20 = 30$$

Number	1	2	3	4	5	6
Observed	20	10	10	20	20	40
Expected	20	20	20	20	20	20
TS	$\frac{(20 - 20)^2}{20}$ = 0	$\frac{(10 - 20)^2}{20}$ = 5	$\frac{(10 - 20)^2}{20}$ = 5	0	0	$\frac{(40 - 20)^2}{20}$ = 20

Critical Value

As with all hypothesis tests, to decide whether this is big or not, **we need to compare it to a Critical Value.**

The table needed is the **Chi-Squared Distribution Table**

(Available on the VLE).

The degree of freedom is $6-1=5$ in this case, and by default sig.lev=**5%**.

(When keeping the sum of the frequencies the same, one can (at most) have free choice over 5 of them, and the last is fixed by those choices.)



Example: throw a die conclusion

TS = 30.

The critical value

CV = 11.070.

TS > CV, so there is a significant difference

We reject the Null and we conclude that
this die is NOT fair.

Chi-Squared Distribution Table

	10%	5%	2.5%	1%	0.5%	0.1%
1	2.706	3.841	5.024	6.635	7.879	10.828
2	4.605	5.991	7.378	9.210	10.597	13.816
3	6.251	7.815	9.348	11.345	12.838	16.266
4	7.779	9.488	11.143	13.277	14.860	18.467
5	9.236	11.070	12.833	15.086	16.750	20.515
6	10.645	12.592	14.449	16.812	18.548	22.458
7	12.017	14.067	16.013	18.475	20.278	24.322
8	13.362	15.507	17.535	20.090	21.955	26.124
9	14.684	16.919	19.023	21.666	23.589	27.877
10	15.987	18.307	20.483	23.209	25.188	29.588
11	17.275	19.675	21.920	24.725	26.757	31.264
12	18.549	21.026	23.337	26.217	28.300	32.909
13	19.812	22.362	24.736	27.688	29.819	34.528
14	21.064	23.685	26.119	29.141	31.319	36.123
15	22.307	24.996	27.488	30.578	32.801	37.697
16	23.542	26.296	28.845	32.000	34.267	39.252
17	24.769	27.587	30.191	33.409	35.718	40.790
18	25.989	28.869	31.526	34.805	37.156	42.312
19	27.204	30.144	32.852	36.191	38.582	43.820

Example: York and “Brexit”

We used a uniform distribution for the die-rolling example, seemed reasonable in the context of that example.

A different example: say we were interested in how many people in an area voted for a particular outcome.

Nationally, the percentage of votes cast is known. Does our area voting pattern match the national voting pattern?

Example: York and “Brexit”

We can look up the results for the local area:

Vote:	Remain	Leave
York	63,617	45,983

These are the observed outcomes

The national result is well known, in percentage terms (*48.1% Remain, 51.9% Leave*), so we split the 109,600 York votes to **obtain the expected**:

Vote:	Remain	Leave
York	63,617	45,983
Expect	52,717.6	56,882.4

And, again, we take the **differences**, **square**, and **divide** by the expected (*in this case, Nation*):

$$TS = \sum \frac{(O - E)^2}{E}$$

This gives us a Test Statistic of **4,342**.

Is it significant? Let's compare it to Critical Value.

Degrees of freedom is $2-1=1$ and CV for 5% significance level is **3.841**.

Example: Accident records

A factory's accident record is analysed. The table below shows the number of accidents which occurred on each day of the week. Are the results consistent with the assumption that accidents are equally likely to occur on any day of the week?

Day	Mon	Tue	Wed.	Thurs	Fri
No. of accidents	17	8	9	7.	19

Step 1. Set up the null and alternative hypotheses

- H_0 : Accidents are equally likely to occur on any day of the week
- H_1 : Accidents are not equally likely to occur on any day of the week. That is, there is an association between the number of accidents and the day of the week

Example: Accident records

Day	Mon	Tue	Wed	Thurs	Fri
No. of accidents	17	8	9	7	19

Step 2. Decide on the significance level (normally, 5%), null distribution, and determine the critical values

What is the critical value?

- Since the total (60) is fixed we are only free to 'choose' values for 4 out of the 5 days. So the *degrees of freedom* is 4
- If we look in the Chi-square table for a 5% (.05) significance level with 4 degrees of freedom we get a critical value of **9.488**

Chi-Squared Distribution Table

	10%	5%	2.5%	1%	0.5%	0.1%
1	2.706	3.841	5.024	6.635	7.879	10.828
2	4.605	5.991	7.378	9.210	10.597	13.816
3	6.251	7.815	9.348	11.345	12.838	16.266
4	7.779	9.488	11.143	13.277	14.860	18.467
5	9.236	11.070	12.833	15.086	16.750	20.515
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Example: Accident records

A factory's accident record is analysed. The table below shows the number of accidents which occurred on each day of the week. Are the results consistent with the assumption that accidents are equally likely to occur on any day of the week?

Day	Mon	Tue	Wed	Thurs	Fri	
No. of accidents	17	8	9	7	19	Total no. of accidents=60
Expected no. of accidents	12	12	12	12	12	

Step 3. Calculate the test statistic

Chi-square (goodness of fit)

$$TS = \sum \frac{(O - E)^2}{E}$$

Example: Accident records

Day	O	E	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Mon	17	12	5	25	2.083
Tues	8	12	-4	16	1.333
Wed	9	12	-3	9	0.750
Thurs	7	12	-5	25	2.083
Fri	19	12	7	49	4.083
Total	60	60			10.332

$$TS = \sum \frac{(O - E)^2}{E}$$

A factory's accident record is analysed. The table below shows the number of accidents which occurred on each day of the week. Are the results consistent with the assumption that accidents are equally likely to occur on any day of the week?

Day	Mon	Tue	Wed	Thurs	Fri
No. of accidents	17	8	9	7	19

Step 4. Decide whether to accept or reject the null hypothesis

Step 1. H_0, H_1

Step 2. $TS=10.332$

Step 3. 5% significance, critical value of 9.488

Step 4.

▪ Since our test statistic (10.332) is greater than CV (9.488) we reject H_0

Step 5. We conclude that there is evidence to suggest that accidents are more likely on particular days

▪ In particular it looks as if Monday and Friday have more accidents than should be expected by chance

Summary

We have seen three applications to check whether frequencies of something match what you would expect.

This is a “goodness of fit” test.

Next, we will see applications to check whether two variables are associated.

This is a test of association.

(We have seen this in the eating breakfast example)

Test of Association

This version is pretty much the same as the previous one, just a little more complicated.

If we're testing for association, (*which is a conditional thing*) then the **null hypothesis is that there is NO correlation** – that the things are completely **independent** of one another
(*in an actual probability sense*).

What do we expect to happen?

Meaning of independence

If two things are independent, remember, we learn NOTHING about one from the other. Then we are back to probability lectures.

If A and B are independent, then

$$P(A \cap B) = P(A) * P(B)$$

For example:

What is the chance that I roll 2 on a die, and flip a head?

$$\frac{1}{6} \times \frac{1}{2} = \frac{1}{12} = 0.083$$



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Example: confidence with numbers

Two variables:

1. I feel very confident with maths (agree, neither, disagree)
2. Gender (male and female)

Is there an association between gender and confidence with numbers?

H_0 : There is no association

H_1 : There is an association

Observed	Disagree	Neither	Agree	Total
Male	8	25	31	64
Female	23	29	43	95
Total	31	54	74	159

Calculate the Critical Value

Chi-Squared Distribution Table

	10%	5%	2.5%	1%	0.5%	0.1%
1	2.706	3.841	5.024	6.635	7.879	10.828
2	4.605	5.991	7.378	9.210	10.597	13.816
3	6.251	7.815	9.348	11.345	12.838	16.266
4	7.779	9.488	11.143	13.277	14.860	18.467
5	9.236	11.070	12.833	15.086	16.750	20.515
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9	14.684	16.919	19.023	21.666	23.589	27.877
10	15.987	18.307	20.483	23.209	25.188	29.588
11	17.275	19.675	21.920	24.725	26.757	31.264
12	18.549	21.026	23.337	26.217	28.300	32.909

In general, data with r rows and c columns will have
 $(r - 1) \times (c - 1)$ degrees of freedom.

This only applies to situations with two variables, though, of course.

So what is the CV if we assume 5% significance?

CV=5.991

Observed	Disagree	Neither	Agree	Total
Male	8	25	31	64
Female	23	29	43	95
Total	31	54	74	159

Calculate the Expected Value

Observed	Disagree	Neither	Agree	Total
Male	8	25	31	64
Female	23	29	43	95
Total	31	54	74	159

40.3%

59.7%

If the overall gender breakdown is 40% / 60%, then for us to learn nothing from Confident with Numbers, those three columns (*Disagree, Neither, Agree*) must also each be in the ratio 40 / 60.

Calculate the Expected Value

So, we split **31**, **54** and **74** in the ratio 40/60:

Expected	Disagree	Neither	Agree	Total
Male			29.6	64
Female			44.4	95
Total	31	54	74	159

40%

60%

If the overall gender breakdown is 40%/60%, then for us to learn nothing from Confident with Numbers, those three columns must also be in the ratio 40/60.

Calculate the Expected Value

So, we split **31**, **54** and **74** in the ratio 40/60:

Expected	Disagree	Neither	Agree	Total
Male	12.4	21.6	29.6	64
Female	18.6	32.4	44.4	95
Total	31	54	74	159

40% → (pointing to 64)

60% → (pointing to 95)

If the overall gender breakdown is 40%/60%, then for us to learn nothing from Confidence with Numbers, those three columns must also be in the ratio 40/60.

So now, whichever column you're in, the gender categories are in the ratio 40/60, and we learn nothing from Confidence with Numbers about Gender.

Calculate the Test Statistic

Observed	Disagree	Neither	Agree	Total
Male	8	25	31	64
Female	23	29	43	95
Total	31	54	74	159

We get the Test Statistic in the same way as before. $\left(TS = \sum \frac{(O-E)^2}{E}\right)$

Expected	Disagree	Neither	Agree	Total
Male	12.4	21.6	29.6	64
Female	18.6	32.4	44.4	95
Total	31	54	74	159

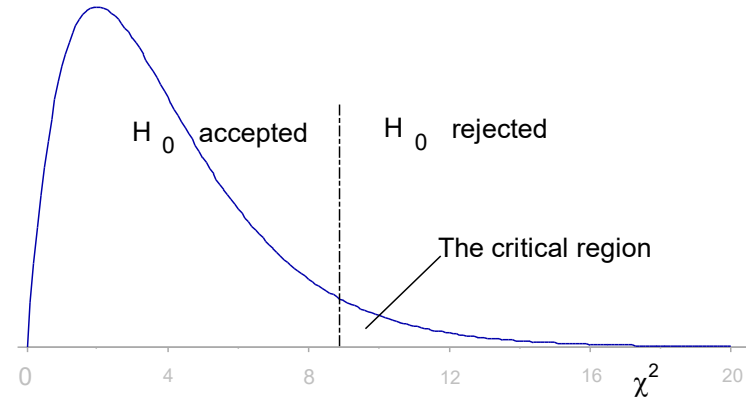
TS	Disagree	Neither	Agree	Total
Male	1.56	0.54	0.06	
Female	1.04	0.35	0.04	
Total				3.60

$$TS = \frac{(8 - 12.4)^2}{12.4} + \frac{(25 - 21.6)^2}{21.6} + 0.06 + 1.04 + 0.35 + 0.04 = 3.60$$

Draw a conclusion

Recall our test statistic was **3.60**.
CV = 5.991 at 5% significance level

TS	Disagree	Neither	Agree	Total
Male	1.56	0.54	0.06	
Female	1.04	0.35	0.04	
Total				3.60



$3.60 < 5.991$, so we fail to reject the null hypothesis.

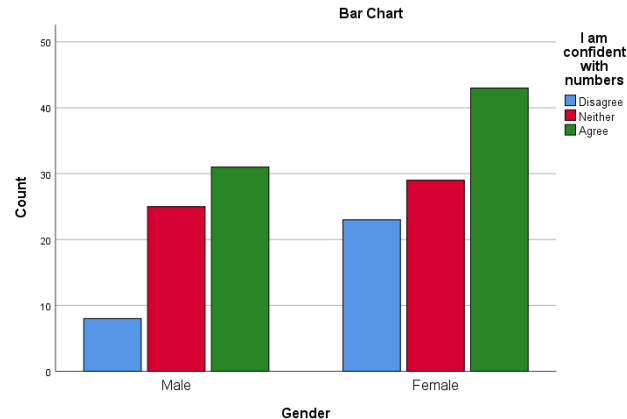
We conclude that Gender and Confidence with Numbers are not significantly associated.

Output - SPSS

Gender * I am confident with numbers
Crosstabulation

Count

		I am confident with numbers			
		Disagree	Neither	Agree	Total
Gender	Male	8	25	31	64
	Female	23	29	43	95
Total		31	54	74	159



Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	3.593 ^a	2	.166
Likelihood Ratio	3.740	2	.154
Linear-by-Linear Association	1.434	1	.231
N of Valid Cases	159		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 12.48.

Exercise: distance to work (1)

The Personnel department of a company want to investigate whether there is any association between the amount of time employees are absent from work and the distance they travel to work.

The table below show the result of a random sample of 100 employee records.

Absence Rate

	Low	Medium	High	Total
Less than 5 miles	21	14	10	45
<u>5 or more miles</u>	<u>14</u>	<u>16</u>	<u>25</u>	<u>55</u>
Total	35	30	35	100

Exercise: distance to work (2)

- **H₀**: No association between absence rate and distance travelled
- **H₁**: There is an association between absence rate and distance travelled

What is the critical value?

The critical value at 5% on 2 degrees of freedom is **5.991**.

	Low	Medium	High	Total
Less than 5 miles	21	14	10	45
5 or more miles	14	16	25	55
Total	35	30	35	100

Exercise: distance to work (3)

	Low	Medium	High	Total
Less than 5 miles	21	14	10	45
5 or more miles	14	16	25	55
Total	35	30	35	100

- If the null hypothesis were true and you picked a person at random what is the probability that they would be in the categories 'low absence rate' and 'travelled less 5 miles'?
- $P(\text{Low}) = 35/100$
- $P(<5 \text{ miles}) = 45/100$
- $P(\text{Low and } <5 \text{ miles}) = \frac{35}{100} \times \frac{45}{100} = 0.1575$
- **Expected value = $\frac{\text{Row total} \times \text{Column total}}{\text{Grand total}} = \frac{35 \times 45}{100}$**

Exercise: distance to work (4)

	Low	Medium	High	Total
Less than 5 miles	21	14	10	45
5 or more miles	14	16	25	55
Total	35	30	35	100

E = $\frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$

O	E	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
21	15.75	5.25	27.5625	1.7500
14	13.5	0.5	0.2500	0.0185
10	15.75	-5.75	33.0625	2.0992
14	19.25	-5.25	27.5625	1.4318
16	16.5	-0.5	0.2500	0.0152
25	19.25	5.75	33.0625	1.7175
				7.032

Exercise: distance to work (Solution)

Step 1. Set up the null and alternative hypotheses

Step 2. Decide on the significance level (normally, 5%), null distribution, and determine the critical values

Step 3. Calculate the test statistic

Step 4. Decide whether to accept or reject the null hypothesis.

Step 5. Draw a conclusion.

Step 1. H_0 : There is no association

H_1 : There is an association

Step 2. The critical value at 5% on Chi-square distribution with 2 degrees of freedom is **5.991**

Step 3. $TS=7.032$

Step 4. Since 7.032 is greater than this value we reject H_0

Step 5. We conclude that there is an association between absence rate and distance travelled

In fact it appears that people who travel further to work have a higher absence rate

Test of Homogeneity

Two variables from **one sample**:

1. I feel very confident with maths
(agree, neither, disagree)
2. Gender (male and female)

Observed	Disagree	Neither	Agree	Total
Male	8	25	31	64
Female	23	29	43	95
Total	31	54	74	159

This is a test of association.

One variable from **two samples**:

1. I feel very confident with maths
(agree, neither, disagree)

Observed	Disagree	Neither	Agree	Total
City I	8	25	31	64
City II	23	29	43	95
Total	31	54	74	159

This is a test of homogeneity.

Sampling methods are different. Procedure of test is the same.

Conclusion

- We introduced yet another way of ‘checking ideas’, i.e. hypothesis testing on categorical data.
- We know how to test goodness-of-fit, test of associations, and test homogeneity.

$$TS = \sum \frac{(O - E)^2}{E}$$

- Next week: **ANOVA** – Taylor Chapter 8